

Disproof of conjecture 00000001682

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Verdict: FALSE

We disprove conjecture 00000001682 as stated. The parenthetical claim of the conjecture — that the generalized Petersen graph $G(n, 2)$ is pancyclic for all $n \geq 11$ — fails already at $n = 11$: the graph $G(11, 2)$ is *triangle-free*, hence cannot be pancyclic because it has 22 vertices and pancyclicity would require a cycle of length 3. The refutation is unconditional, elementary, and machine-checked (see `reproduce.py` and the Lean 4 project under `lean4/`).

1 The conjecture

Quoted verbatim from `conjectures/00000001682.md`:

English. Conjecture: The pancyclicity threshold of generalized Petersen graphs $G(n, 2)$ has a complete classification for $n \geq 11$ ($G(n, 2)$ pancyclic).

中文。猜想: 广义 Petersen 图 $G(n, 2)$ 的满圈泛环性 (pancyclic) 阈值为 $n \geq 11$ 的完全分类 ($G(n, 2)$ 泛环)。

We first fix the two objects appearing in the statement.

Definition 1 (Generalized Petersen graph). For integers $n \geq 3$ and $1 \leq k \leq (n - 1)/2$, the generalized Petersen graph $\text{GP}(n, k)$ has vertex set

$$V = \{u_0, u_1, \dots, u_{n-1}\} \cup \{v_0, v_1, \dots, v_{n-1}\}, \quad |V| = 2n,$$

and edge set

$$E = \{u_i u_{i+1}\} \cup \{u_i v_i\} \cup \{v_i v_{i+k}\},$$

where all subscripts are read modulo n . The edges $u_i u_{i+1}$ are the *outer* edges, the edges $u_i v_i$ the *spokes*, and the edges $v_i v_{i+k}$ the *inner* edges. The conjecture concerns the case $k = 2$.

Definition 2 (Pancyclic). A graph G is *pancyclic* if it contains a simple cycle of every length L with $3 \leq L \leq |V(G)|$. Thus $\text{GP}(n, 2)$ is pancyclic exactly when it has a cycle of each length $3, 4, \dots, 2n$.

The conjecture therefore asserts that $\text{GP}(n, 2)$ is pancyclic for every $n \geq 11$. It is this assertion that we refute.

2 The witness $G(11, 2)$ is triangle-free

2.1 Proof by the neighbourhood structure

Indices are taken modulo n . From Definition 1 with $k = 2$, the neighbourhoods in $GP(n, 2)$ are

$$N(u_i) = \{u_{i-1}, u_{i+1}, v_i\}, \quad N(v_i) = \{v_{i-2}, v_{i+2}, u_i\}. \quad (1)$$

Theorem 3. *The graph $GP(11, 2)$ contains no triangle.*

Proof. Let $U = \{u_0, \dots, u_{10}\}$ and $W = \{v_0, \dots, v_{10}\}$.

No triangle inside U . By (1), on U the graph is the cycle $u_0 u_1 \dots u_{10} u_0$ of length 11. A cycle of length 11 is triangle-free (as $11 \geq 4$).

No triangle inside W . By (1), on W the graph joins v_i to $v_{i\pm 2}$. Since $\gcd(11, 2) = 1$, the map $i \mapsto 2i$ is a bijection of $\mathbb{Z}_{11}$, so the inner graph is a single cycle $v_0 v_2 v_4 \dots v_9 v_{11} v_0$ (all subscripts mod 11) of length 11, hence triangle-free.

No mixed triangle. A triangle meeting both U and W either contains a vertex $v_i \in W$ together with two distinct vertices of U , or contains a vertex $u_i \in U$ together with two distinct vertices of W (a triangle has three vertices and only two classes, so some class occurs twice). In the first case v_i would need two neighbours in U , but (1) gives $|N(v_i) \cap U| = 1$, namely u_i . In the second case u_i would need two neighbours in W , but $|N(u_i) \cap W| = 1$, namely v_i . Both are impossible.

Every triangle of $GP(11, 2)$ is therefore monochromatic (all three vertices in U , or all three in W), and neither case occurs. Hence $GP(11, 2)$ is triangle-free. $\square$

A completely explicit variant: evaluating (1) for all $\binom{22}{3} = 1540$ vertex triples of $GP(11, 2)$ yields zero triangles. This finite enumeration is carried out by `reproduce.py` and is certified by the kernel-reduction theorem `gp11_triangle_count` in the Lean 4 formalisation.

2.2 Why $n = 11$ is special: triangles occur only for $n = 6$

The same argument classifies all triangles of $GP(n, 2)$.

Proposition 4. *Let $n \geq 4$. Then $GP(n, 2)$ contains a triangle if and only if $n = 6$. In particular $GP(n, 2)$ has a triangle precisely when 6 is divisible by n among $\{1, 2, 3, 6\}$, i.e. the triangle exists only for the degenerate $n \in \{1, 2, 3\}$ and for $n = 6$.*

Proof. As in the proof of Theorem 3, no triangle can be mixed: v_i has exactly one neighbour u_i in the outer layer and u_i has exactly one neighbour v_i in the inner layer, by (1). So a triangle is either fully outer or fully inner.

Outer triangles. The outer layer is the cycle C_n ; it has a triangle iff $n = 3$.

Inner triangles. The inner layer joins v_i to $v_{i\pm 2}$. If n is odd then $\gcd(n, 2) = 1$ and it is a single cycle C_n , which has a triangle iff $n = 3$. If n is even then it is the disjoint union of two cycles $C_{n/2}$, which has a triangle iff $n/2 = 3$, i.e. $n = 6$.

Combining, for $n \geq 4$ the only possible triangle occurs in the inner layer at $n = 6$. There the inner layer is two triangles, so $GP(6, 2)$ has exactly two triangles. The brute force confirms both facts: the triangle counts for $n = 5, 6, 12$ are 0, 2, 0 respectively. $\square$

Remark 5. In particular $\text{GP}(11, 2)$ is triangle-free, as are $\text{GP}(n, 2)$ for all $n \geq 4$ with $n \neq 6$. Thus for every $n \geq 7$ the graph $\text{GP}(n, 2)$ is automatically *not* pancyclic, regardless of its other cycle lengths.

3 The witness is not pancyclic, so the conjecture is false

Corollary 6. *The graph $\text{GP}(11, 2)$ is not pancyclic.*

Proof. By Theorem 3 it has no 3-cycle. Since $|V(\text{GP}(11, 2))| = 22$, pancyclicity would require a cycle of length 3 (Definition 2), which does not exist. $\square$

Corollary 7. *Conjecture 00000001682 is false. Its parenthetical claim “ $\text{GP}(n, 2)$ pancyclic” for $n \geq 11$ fails at $n = 11$.*

Indeed an exhaustive search shows that $\text{GP}(11, 2)$ is missing the cycle lengths

$$3, 4, 6, 7, 22$$

and possesses cycles of all of the other lengths $5, 8, 9, \dots, 21$. The two smallest missing lengths 3 and 4 already rule out pancyclicity. The absence of a Hamiltonian (22-)cycle is consistent with the fact that $\text{GP}(11, 2)$ is hypohamiltonian: $11 \equiv 5 \pmod{6}$, and the known classification for these graphs is a *Hamiltonicity* classification, not a pancyclicity one (see Section 5).

4 The claim fails for every $n = 5, \dots, 16$

We computed, by exhaustive search, the exact set of lengths $L \in \{3, \dots, 2n\}$ for which $\text{GP}(n, 2)$ has no cycle. Every one of these graphs misses at least one length, so none is pancyclic.

n	$ V = 2n$	triangles	missing cycle lengths
5	10	0	3, 4, 7, 10
6	12	2	4
7	14	0	3, 4
8	16	0	3, 6
9	18	0	3, 4, 6
10	20	0	3, 4, 6, 7, 19
11	22	0	3, 4, 6, 7, 22
12	24	0	3, 4, 7
13	26	0	3, 4, 6, 7
14	28	0	3, 4, 6
15	30	0	3, 4, 6, 7
16	32	0	3, 4, 6, 7, 31

Corollary 8. *$\text{GP}(n, 2)$ is not pancyclic for any $n \in \{5, 6, \dots, 16\}$. In particular the asserted threshold “pancyclic for $n \geq 11$ ” is false, and no classification of a pancyclicity threshold of the stated form can hold.*

For $n = 5$ the graph is the Petersen graph, which is well known to be non-Hamiltonian and to lack cycles of length 3 and 4; the table adds the missing length 7 and the Hamiltonian length 10. For $n = 11$ the missing Hamiltonian length 22 reflects hypohamiltonicity.

5 The structural reason, and what the true classification is

The example is not an isolated accident: for every $n \geq 7$, $\text{GP}(n, 2)$ is triangle-free by Proposition 4 and therefore not pancyclic. The property that actually undergoes a threshold-type classification for $\text{GP}(n, 2)$ is *Hamiltonicity*, not pancyclicity:

- $\text{GP}(n, 2)$ is hypohamiltonian (non-Hamiltonian, but deleting any vertex leaves a Hamiltonian graph) for $n \equiv 5 \pmod{6}$, e.g. for $n = 5, 11$;
- $\text{GP}(n, 2)$ is Hamiltonian for the other values of n in the range we tested, e.g. for $n = 6, 7, 8, 9, 10$ and for $12 \leq n \leq 16$.

This is consistent with the exhaustive data: the only tested graphs without a Hamiltonian cycle are $n = 5$ and $n = 11$, both $\equiv 5 \pmod{6}$. The conjecture under consideration appears to be a garbled version of that Hamiltonicity classification, transported to the wrong graph property (pancyclicity, which fails for trivial girth reasons).

6 Reproducibility

The brute-force checks use only the Python 3 standard library:

```
$ python3 reproduce.py
... (per- $n$  table of missing cycle lengths, witness checks)
PASS: all checks verified; conjecture 00000001682 is FALSE.
```

The script constructs $\text{GP}(n, 2)$ for $n = 5, \dots, 16$, counts triangles over all $\binom{2n}{3}$ triples, and decides cycle existence for every length $L = 3, \dots, 2n$ by two exhaustive methods: enumeration of all $\binom{2n}{L}$ vertex subsets (for small L , testing Hamiltonicity of the induced subgraph) and a pruned DFS cycle search (for larger L). It asserts that $\text{GP}(11, 2)$ has no triangle, prints the table, and exits non-zero on any failure. The LaTeX document compiles with `tectonic main.tex`; the PDF is `build/main.pdf`.

7 Formalisation

The witness is formalised in the accompanying Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). Vertices of $\text{GP}(11, 2)$ are modelled as $\text{Fin } 2 \times \text{Fin } 11$ with $u_i = (0, i)$ and $v_i = (1, i)$; the adjacency relation is a decidable `Bool` predicate implementing Definition 1 with all indices modulo 11; a triangle is three pairwise adjacent vertices; and a cycle of length L is an injective closed walk with L vertices, giving

$$\text{HasCycleOfLength } G \ L, \quad \text{Pancyclic } G := \forall L, 3 \leq L \rightarrow L \leq 22 \rightarrow \text{HasCycleOfLength } G \ L.$$

The main statements are

```
theorem gp11_triangle_free (a b c : Fin 2) (i j k : Fin 11) :
  ¬ Triangle (gp 11 2) (a, i) (b, j) (c, k)
theorem gp11_triangle_count : triangleCount 11 2 = 0
theorem gp11_no_3cycle : ¬ HasCycleOfLength (gp 11 2) 3
theorem gp11_not_pancyclic : ¬ Pancyclic (gp 11 2)
theorem conjecture_00000001682_false :
  (∀ x y z, ¬ Triangle (gp 11 2) x y z) ∧ ¬ Pancyclic (gp 11 2)
```

Triangle-freeness is discharged by kernel reduction over all $2 \cdot 2 \cdot 2 \cdot 11 \cdot 11 \cdot 11$ layer/index choices, and independently by the exhaustive count `gp11_triangle_count = 0`. From `Pancyclic` one extracts a 3-cycle, which is a triangle, contradicting triangle-freeness. The axiom audit (`lake env lean Check.lean`) reports no `sorryAx`; the proofs depend only on the core axioms `propext` and `Quot.sound`.